ITA1441 – ETHICAL HACKING

LAB RECORD – EXPERIMENTS 11 TO 13

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EXPERIMENT 11: PACKET ANALYSER TOOL

Experiment question: Capture, filter, and inspect network packets using a packet analyser tool.

Aim

To capture and examine network packets using Wireshark.

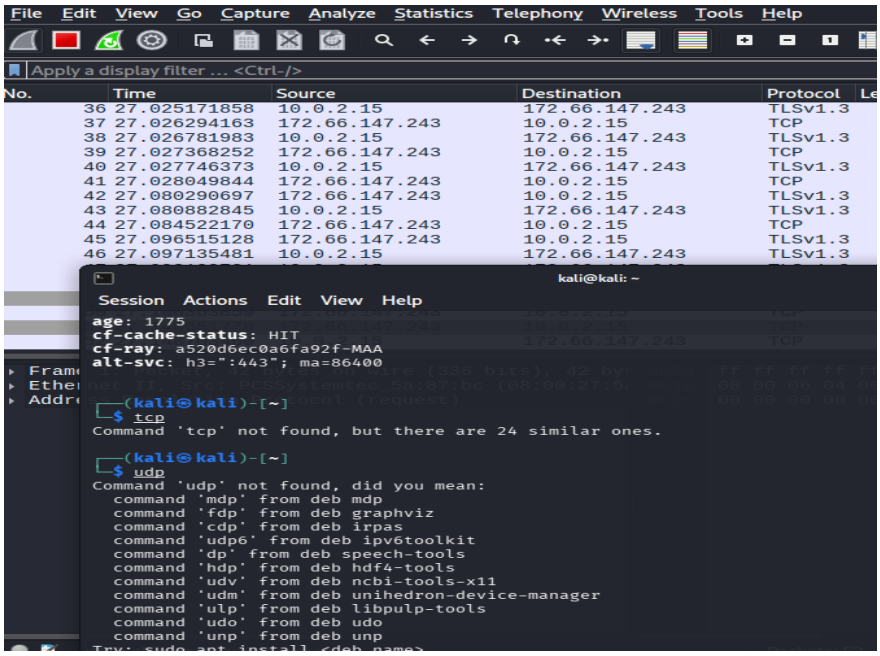
Algorithm

1. Open Wireshark in Kali Linux.
2. Start capturing on the active network interface, eth0.
3. Generate ICMP traffic by running a ping command to 8.8.8.8.
4. Observe ICMP echo request and reply packets in Wireshark.
5. Stop the capture and save the screenshot.

Command(s) used

```
ping -c 4 8.8.8.8
Wireshark display filter: icmp
```

Output Screenshot



Result: Wireshark displayed captured ICMP echo request and reply packets between the Kali interface (10.0.2.15) and 8.8.8.8.

EXPERIMENT 12: PORT SCANNING TOOLS

Experiment question: Perform port scanning using Nmap scan techniques.

Aim

To explore TCP connect, TCP SYN, UDP, and TCP ACK scanning options using Nmap against the local host.

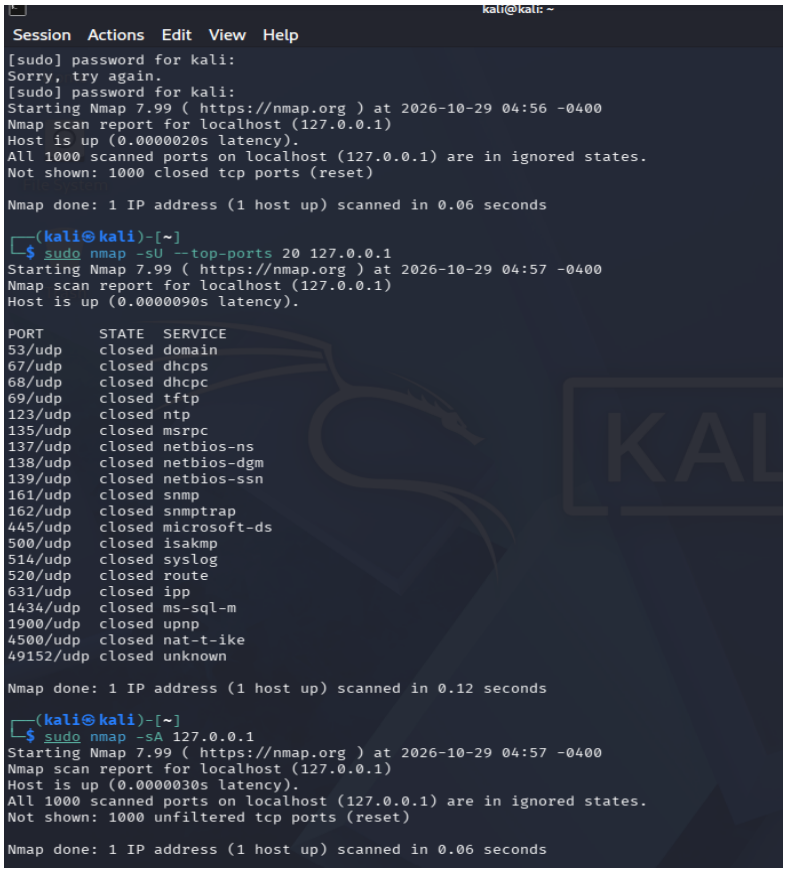
Algorithm

6. Open Kali Linux Terminal.
7. Run a TCP connect scan against 127.0.0.1.
8. Run a TCP SYN scan using sudo and authenticate when prompted.
9. Run a UDP scan of the 20 most common ports.
10. Run a TCP ACK scan and observe the reported port states.
11. Capture the terminal output.

Command(s) used

```
nmap -sT 127.0.0.1
sudo nmap -sS 127.0.0.1
sudo nmap -sU --top-ports 20 127.0.0.1
sudo nmap -sA 127.0.0.1
```

Output Screenshot



Result

The TCP SYN scan showed the local host was up with the scanned TCP ports closed. The UDP scan reported the selected UDP ports as closed. The TCP ACK scan reported the scanned TCP ports as unfiltered.

EXPERIMENT 13: NMAP TIMING AND PERFORMANCE / HOST DISCOVERY

Experiment question: Explore Nmap host-discovery and scan options.

Aim

To observe Nmap options for host discovery, skipping discovery, and disabling DNS resolution.

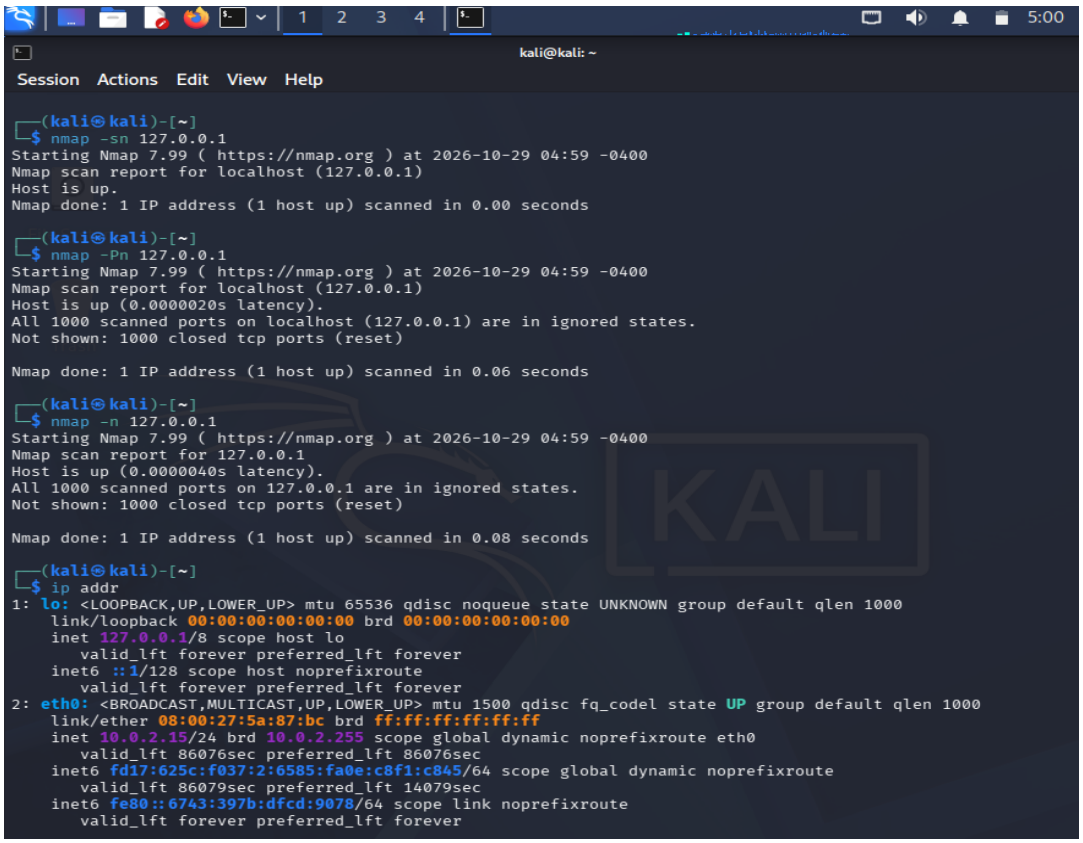
Algorithm

12. Open Kali Linux Terminal.
13. Run a host-discovery scan against 127.0.0.1 using -sn.
14. Run a scan with -Pn to skip host discovery.
15. Run a scan with -n to disable DNS resolution.
16. Use ip addr to view the local network interfaces and addresses.
17. Capture the terminal output.

Command(s) used

```
nmap -sn 127.0.0.1
nmap -Pn 127.0.0.1
nmap -n 127.0.0.1
ip addr
```

Output Screenshot



Result: Nmap reported that localhost was up. The -Pn and -n scans completed against 127.0.0.1, and ip addr displayed loopback and eth0 interfaces, including eth0 address 10.0.2.15.